Solidity OOPs

Part - 5

Contents

Contract and Objects

Inheritance

Abstract Class

Interface

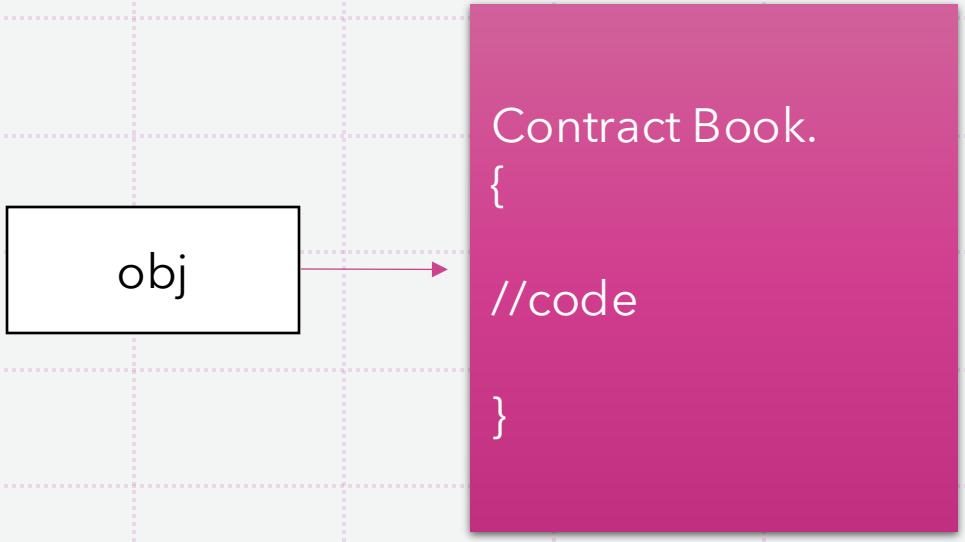
Contract and Object

- Solidity Contracts are like a class in any other object-oriented programming language.
- A class is a description of an object's property and behavior.
- Object is a real world entity.

Code Eater
Code Eater

Contract and Object

Book obj = new Book( );



Note - Constructor of a contract is called as soon as its contract is created.

Code Eater

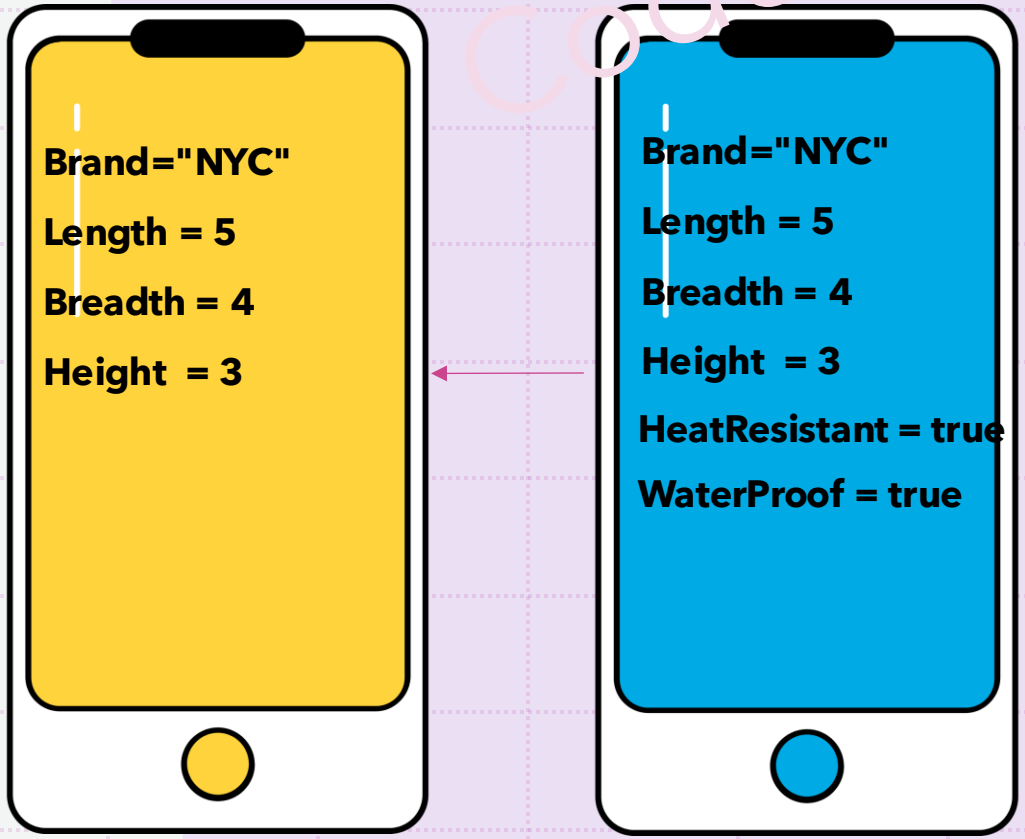
```
contract Book {
    uint length;
    uint breadth;
    uint height;

    function setDimension(uint _length, uint _breadth, uint _height) public {
        length = _length;
        breadth = _breadth;
        height = _height;
    }

    function getDimension() public view returns (uint, uint, uint) {     infinite ga
        return (length, breadth, height);
    }
}

contract D {
    Book obj = new Book();

    function getInstance() public view returns (Book) {     infinite gas
        return obj;
    }

    function writeDimension(uint _length, uint _breadth, uint _height) public {
        obj.setDimension(_length, _breadth, _height);
    }

    function readDimension() public view returns (uint, uint, uint) {     infinite g
        return obj.getDimension();
    }
}
```

Code - [Click Here]

Code Eater

Inheritance

1. Inheritance in Solidity is **the procedure in which one contract inherits the attributes and methods of another contract**.



Code Eater

```
contract Car {
    uint public wheels = 4;
    uint public doors = 4;
    string public brandName = "CTE";
    uint public headlights = 2;
    bool public safetyBag = true;
}

contract SuperCar is Car {
    uint public speed = 400;
    uint public modelNumber = 121;
    string public modelName = "Texxo";
}
```

Code - Click Here

Code Eater
Code Eater

Abstract Contracts

Code Eater

- **Abstract contracts** are contracts that can have functions without its implementation.
- To make a contract **abstract** you have to use abstract keyword.
- Function without implementation must contain **virtual** keyword.
- The abstract contract defines the structure of the contract and any derived contract inherited from it should provide an implementation for the incomplete functions. And that functions should contain **override** keyword.
- And if the **derived contract** is also not implementing the incomplete functions then that derived contract will also be **marked as abstract**.
- Contract marked as **abstract** contract cannot be deployed.

```
pragma solidity >=0.7.0 <0.9.0;

abstract contract Parent {
    string public str;
    address public manager;

    constructor() {    - gas - gas
        str = "Hello World";
        manager = msg.sender;
    }

    function setter(string memory _str) public virtual;

}

contract Child is Parent {
    uint public x;

    function setter(string memory _str) public override {
        str = _str;
    }
}
```

Code - [Click Here]

Code Eater
Code Eater

Interface

Code Eater

- An interface is an agreement or a contract between itself and any contract that implements it.
- **Interfaces restrictions** -
  - They can only inherit from other interfaces but not from other contracts.
  - They cannot declare state variables.
  - They cannot declare constructor.
  - Functions can be declared but not implemented. All declared functions must be **external**.

Interface

```
// Define the interface
interface IGreeter {
    function greet() external view returns (string memory);    - gas
}

// Implement the interface in a contract
contract Greeter is IGreeter {
    function greet() external pure override returns (string memory) {    infinite ga
        return "Hello, world!";
    }
}

// A contract that interacts with the Greeter contract through the interface
contract GreeterUser {
    function getGreeting(address _greeter) external view returns (string memory) {
        return IGreeter(_greeter).greet();
    }
}
```

Code

Code Eater

Interface

- Aur Jyada restrictions
- function ko unimplemented form me likh sakte ho. Nothing else.
- Agar aap ek interface ko inherit karoge to apka contract usse interface ke rules ke hisab se chalega.